Roberto Hernandez Cantoral

+1(970)-730-7392 | cantoral.roberto9@gmail.com | LinkedIn

Education

University of Colorado Boulder **May 2026**
Bachelor of Science in Mechanical Engineering GPA: *3.40/4.0*

Work Experience

Mechanical Engineering Intern | Citrine LLC | Lakewood, CO **Summer 2025**

- Built HVAC energy models in TRACE 3D Plus, including climate-zone setup and heating/cooling load calculations; ran life-cycle cost analysis comparing three proposed systems.
- Improved water piping sizing calculations by designing a custom Excel plumbing calculator using VLOOKUP, Conditional Control-Flow structure, and Boolean logic to automate GPM, WSFU, CWFU, DFU, and HWFU calculations based on IPC Appendix E and Chapter 7 standards.
- Modeled mechanical and plumbing systems in Autodesk Revit (BIM), implementing ductwork supply and return air systems (VAV, AHU), sanitary plumbing (PVC, Wye connections, P-traps), and copper hot/cold water lines with full fixture and equipment schedule documentation.

Exhibit Coordinator | EUREKA! Science Museum | Grand Junction, CO **Aug 2023 – May 2026**

- Designed and developed hands-on interactive exhibits, integrating principles of physics, mathematics, and engineering to create engaging visitor experiences and increase visitor attendance by 30%.
- Contributed to the maintenance and repair of museum exhibits, diagnosed issues, and applied problem-solving skills to maintain optimal functionality and a high-quality visitor experience.

Projects

Smart Glasses for people with Homonymous Hemianopia | *Senior Design Capstone Project* **Spring 2026**

- Designed and fabricated a wearable AI-enabled assistive device frame using OnShape with custom 3D-printed PLA, and structurally validated with SolidWorks FEA simulation.
- Co-developed to integrate a Raspberry Pi Zero 2 W and AI camera running YOLO object detection to identify blind-side hazards in real time, projecting text alerts via a micro-OLED display.
- Engineered an adjustable optical mount with independent beam-splitter angle and micro-OLED focus control to accommodate individual user visual geometry.
- Validated 20+ object types in lab and field tests; delivered at $1,700/unit, 83% under budget.

Quality Control Conveyor System | *ENGR 435 – Industrial Controls* **Spring 2026**

- Automated a quality control conveyor with a Micro850 PLC to detect, sort, and track good/bad parts.
- Wrote ladder logic using photoelectric/inductive sensors to meter parts and trigger pneumatic rejection.
- Integrated E-stop, 30% jog mode, and status indicator lights for safe operator control.
- Configured an HMI to set motor speed and log real-time counts and first-pass yield for 15+ parts.

Modular Automated Watering System | *First Year Engineering Project* **Spring 2024**

- Programmed the ESP82 microcontroller with Arduino IDE to compare soil moisture readings against a set threshold and automated the water pump operation to maintain optimal moisture levels.
- Designed automated irrigation to meet species-specific plant needs across a 2-month watering period.
- Created 3D models and drawings using SOLIDWORKS, selecting appropriate pump, sensors, and materials.
- Utilized Manual Mill and 3D Printing to fabricate key components, including a water manifold and an electronic sheltering box, to optimize water distribution and protect electronics from environmental damage.

Activities and Awards

- Society of Hispanic Professional Engineers (SHPE) member
- Won the Computer Science & Engineering – First Year Design Project, Spring 2024
- Colorado Mesa University's Men's Soccer team (NCAA – Division II)

Skills

Software: Python, C, Arduino IDE Programming, JavaScript, Excel, SolidWorks, Onshape, MATLAB, Ladder Logic, HMI Development, Autodesk Revit (BIM), TRACE 3D+

Hardware: Microcontrollers, Sensors, DC Motors, Water Pumps, Pneumatic Cylinders, Micro-OLED, Micro850 PLC

Manufacturing: Lean manufacturing, 3D printing, Six Sigma, Soldering, Manual Mill, Lathe